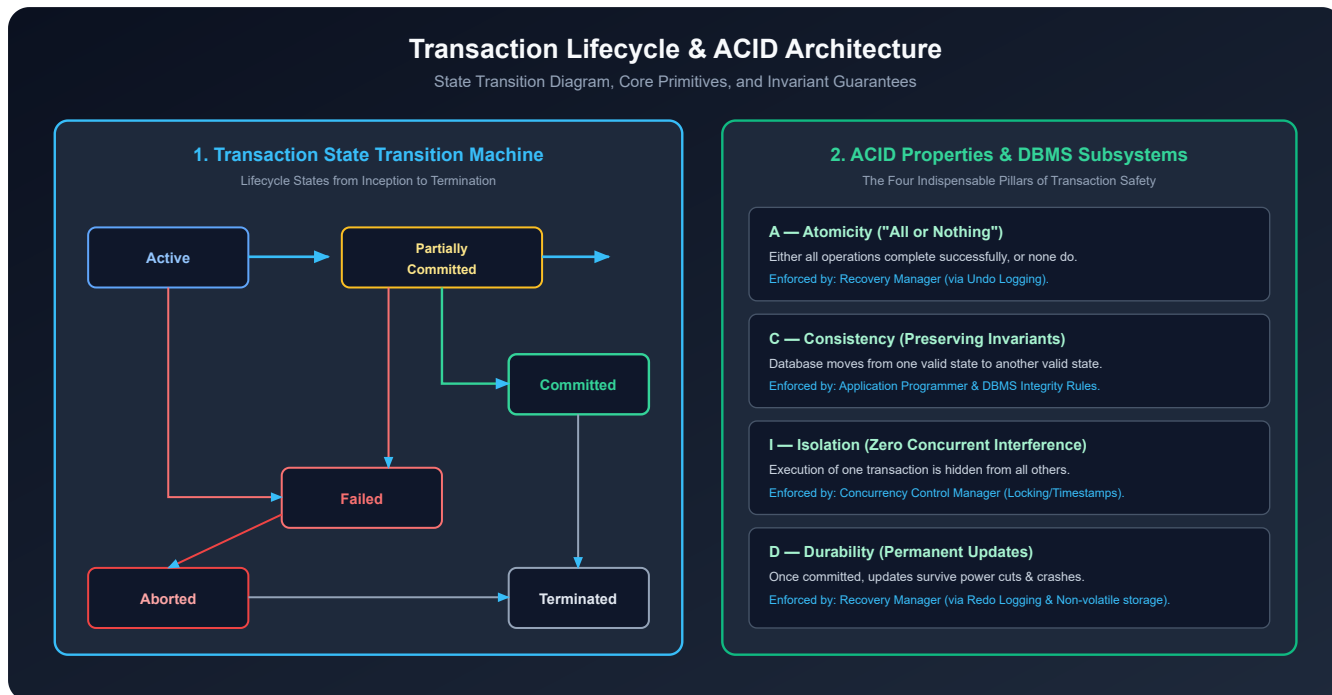


Module 01: Transaction Concepts & ACID Properties



> **Course:** Database Management Systems (AKTU Code: **BCS501**)

****Topic Focus:**** Transaction Definition, Atomic Operations ('Read'/'Write'), Transaction State Transition Machine, ACID Properties & Their DBMS Enforcement Subsystems, Failure Types.

1. What is a Transaction?

Database Management Systems me ek **Transaction** logically related operations ka ek single collection (unit of work) hota hai jo ya toh poora execute hota hai ya bilkul bhi nahi.

Formal Definition: A transaction is a single logical unit of database processing that accesses and possibly updates various data items.

Real-world example: **Bank Fund Transfer**

- Account \$A\$ se Account \$B\$ me \ \$500 transfer karna:

1. Read(A)
2. $A = A - 500$
3. Write(A)
4. Read(B)
5. $B = B + 500$
6. Write(B)
7. Commit

- Agar step 3 ke baad power chali jaye ya server crash ho jaye, toh system ko aisi state me nahi chhod sakte jahan \$A\$ se paise cut gaye par \$B\$ ko nahi mile!

2. Fundamental Operations in a Transaction

Har high-level database query physically do primitive operations me translate hoti hai:

1. Read(X):

- Disk/Database se data item \$X\$ ko fetch karke transaction ke local memory buffer me copy karta hai.

2. Write(X):

- Transaction ke local memory buffer se updated value ko database buffer/disk par write-back karta hai.

3. Transaction State Transition Machine

Ek transaction apne execution lifecycle ke dauran specified formal states se guzarta hai:

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Terminated

State Explanations:

- 1. Active:** Initial state jahan transaction start hota hai aur operations execute kar raha hota hai.
- 2. Partially Committed:** Transaction ka aakhiri instruction execute ho chuka hai, par updates abhi bhi main memory buffers me hain (disk par permanently flush nahi hue).
- 3. Committed:** Transaction ke sabhi updates permanently non-volatile storage (disk) par save ho chuke hain. Ab transaction ko rollback nahi kiya ja sakta.
- 4. Failed:** Normal execution ke waqt agar hardware crash, logical bug, ya system error aaye toh transaction Failed state me chala jaata hai.
- 5. Aborted:** Failed transaction ke saare changes undo (rollback) karke database ko original initial consistent state me restore kiya jaata hai.
- 6. Terminated:** Transaction system chhod kar exit kar chuka hai (either successfully committed or cleanly aborted).

4. ACID Properties & DBMS Enforcement Architecture

Database consistency maintain karne ke liye Jim Gray dwara define kiye gaye **ACID Properties** mandatory hain:

Property	Core Guarantee	Real-World Scenario	Responsible DBMS Subsystem
Atomicity	***"All or Nothing"*** — Ya toh saare steps execute honge ya ek bhi nahi.	Agar paisa debit hua toh credit zaroor hoga; crash hone par pura debit undo ho jayega.	**Recovery Manager** (Undo logging)
Consistency	**Preserving Invariants** — Database transaction se pehle aur baad me legal valid rules satisfy karega.	Total sum of money $$(A + B)$$ transfer ke pehle aur baad exactly same rahegi.	**Application Developer** + **Integrity Subsystem**
Isolation	**Concurrent Independence** — Do parallel transactions ek doosre ke intermediate states ko dekh nahi sakte.	User $$(T_1)$$ ke debit hone ke turant baad par credit se pehle User $$(T_2)$$ balance check kare toh use intermediate galat data na dikhe.	**Concurrency Control Manager** (Locks & Timestamps)
Durability	**Permanence** — Commit message aane ke baad system crash ho jaye toh bhi data kabhi kho nahi sakta.	Flight ticket book ho gayi toh server restart hone par bhi seat booked hi rahegi.	**Recovery Manager** (Redo logging & WAL)

5. Architectural Diagram

6. Solved AKTU Examination Question (10 Marks)

Question: Explain the ACID properties of a transaction. What are the various states through which a transaction passes during its lifecycle? Explain with a neat state transition diagram. *(AKTU 2021-22, 2023-24)*

Model Answer Summary:

1. State definition of Transaction as single logical unit of work with Bank transfer example.
2. Draw the 6-state transition diagram (Active, Partially Committed, Committed, Failed, Aborted, Terminated).
3. Explain conditions for transitioning between states:

- Partial Commit \$

ightarrow\$ Commit: When log records and dirty pages are safely written to non-volatile disk.

- Active \$

ightarrow\$ Failed: System error, divide-by-zero, power loss.

- Failed \$

ightarrow\$ Aborted: Rollback execution restoring before-images from undo log.

4. Tabulate ACID properties with their respective DBMS manager components.